Print and Document Services

Pages 367-408

Printing Terminology

- Print Server
  - A central device that accepts incoming requests for printing and submits them to the appropriate printing device
- Print Job
  - A document or item to be printed
- Spooling
  - Preparing a print job for submission to the physical print device by converting a document format and storing it on the printer.
  - In Windows, spooling is performed by the Print Spooler Service
- Queue
  - The waiting area where spooled devices are kept while waiting to be sent to physical printer
- Print Driver
  - The translator that takes a spooled item and formats it in a language understood by the physical printer
- Printer
  - The logical object that creates a print job.
- Print Device
  - The Physical object that holds paper and ink/toner.

Managing Printers in Windows

Printers can be managed locally in the following ways:

- Devices and Printers applet (control panel)
  OR
  Settings → Devices → Printers and Scanners
  - Can be used to add a **local** or **network** printer, view the print queue, set a printer as default, specify printing preferences and settings, configure properties of the printer, and delete a printer.

The printing Process (Print Spooler; Local)

1. The application invokes a printing method and begins to prepare the document for printing.
2. The application stores the document in the spool folder (C:\Windows\System32\spool\PRINTERS).
3. It is formatted using either Enhanced Metafile (EMF) or XML Paper Specification (XPS)
4. The document in the spool folder is converted to a format the printer can accept using instructions in the print driver. This is called *rendering*. At this point, the document has been converted into a print job.
5. The print job is sent to the locally attached or network-attached printer.
6. The print *device* prints the document.

XPS (XML Paper Specification)

- A page description language (document format) that describes how a page should be displayed (supported in Server 2016)
  - .XPS file association.
- Supported natively in Windows.
- Competitor to Adobe's PDF specification (but not really as XPS has limited support outside of Windows).
- XPS formatted documents will look *identical* once they are printed on a page, while documents formatted for Word or other document imaging programs are not formatted for the paper, they are formatted for the screen, so they make look slightly different once printed.
- Windows includes an XPS viewer which allows you to view XPS documents, configure Active Directory rights management or apply a digital signature

Managing Printers in Windows

Network Printers are connected to in the following ways:

- Directly (using the IP address). This is called Direct Printing.
- Using SMB and the share path
  - Can be a printer shared by a client or a server
  - Uses UNC path \\Server\Printer
  - Or a printer shared by a print server (often listed in AD to help find these printers). These printers are often deployed using GPO.
- Print Management console
  - A management console designed for enterprise printing environments, which allows you to manage print servers across a network from a single workstation
  - Available only as part of the Print and Document Services role

The printing Process (Print Spooler; Shared)

1. The application invokes a printing method and begins to prepare the document for printing.
2. The application stores the document in the spool folder (C:\Windows\System32\spool\PRINTERS).
3. It is formatted using either Enhanced Metafile (EMF) or XML Paper Specification (XPS)
4. The print job is sent from the print client to the print server across the network using the correct protocol (SMB, IPP, or LPD).
5. The print server stores the print job within a spool folder under C:\Windows\System32\spool\PRINTERS.
6. The print server sends the print job to the locally attached or network-attached print device after the print device acknowledges that it is ready to receive it.
7. The print device proceeds to print the document and notifies the print server that the print job has completed.
8. The print server notifies the print client that the print job has completed.
9. The print client notifies the user that the print job has completed.

Managing Print Documents

- To access the print queue, right-click a printer and click "See what's printing"
- You can manage each document by right-clicking the document in the queue and using one of the following options:
  - Pause
  - Resume
  - Restart
  - Cancel
  - Properties

Managing Print Documents and Printers

- You can manage print documents with PowerShell cmdlets, too.
- `Get-PrintJob`
  - Same as accessing a document's Properties
- `Suspend-PrintJob`
  - Same as pausing a document
- `Resume-PrintJob`
- `Restart-PrintJob`
- `Remove-PrintJob`
  - Same as deleting a document.
- `Get-Printer`
  - Shows installed printers
- `Get-PrinterConfiguration`
  - Shows configurations of selected printer.

Printer Servers in Windows

Print and Document Services Role

- Installing the Print Services role on a server provides additional tools that are particularly helpful to administrators involved with network printing on an enterprise scale.

Do this…

- On Server2
  - Add Roles and Features
  - Printer and Document Services
  - Restart (if required)

Role Services for the Print Services Role

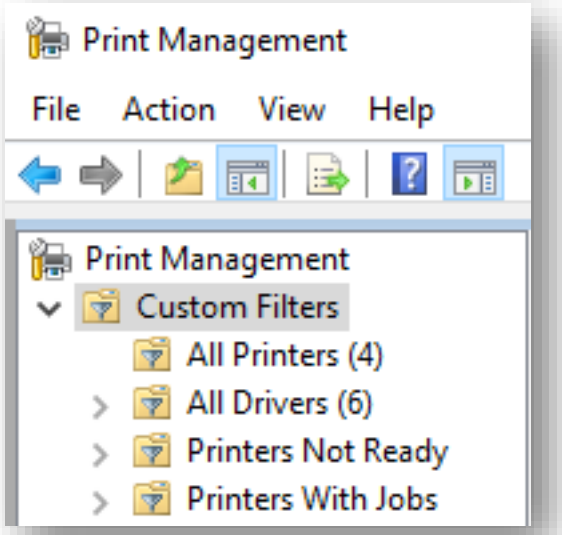
- Print Server (*Required*)
  - Installs the Print Management console, which enables administrators to deploy, monitor, and manage printers throughout the enterprise (to and from other servers).
- LPD Service
  - Adds the TCP/IP Print Server service using Line Printer Daemon (LPD) (an alternative to the print spooler)
  - UNIX clients running the Line Printer Remote (LPR) program to send print jobs to Windows printers.

Do this…

- Install the HP Universal Printing PCL 6 Driver using the installer. (the driver is located in BS)
- Name : Default
- Share as HP4550 A
- Note: Print Test Page

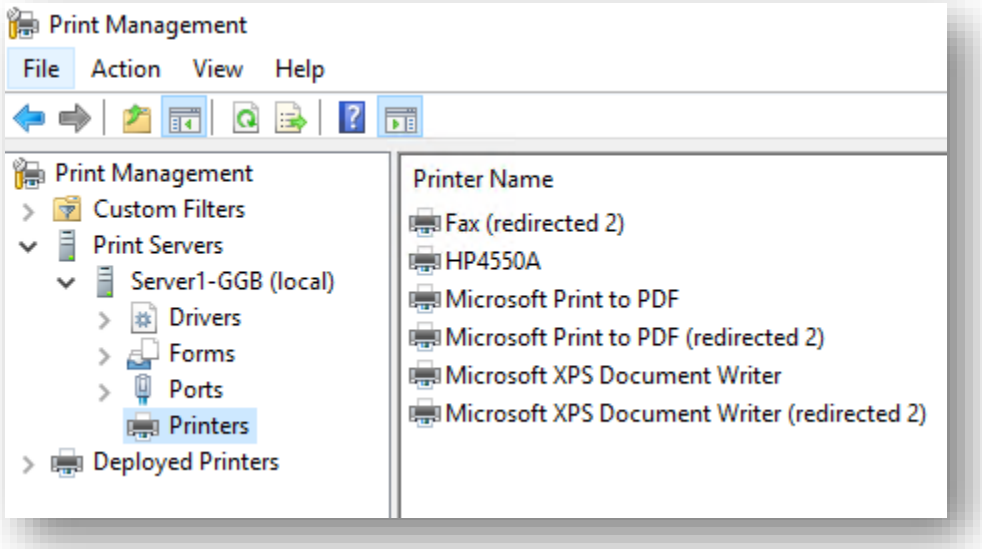
Print Management Console

- Each print Server can be managed VIA the Print Management MMC snap-in.
- All Printers – This filter shows each printer installed on the print server.
- All Drivers – All drivers for all printers installed on the server
- Printers not Ready – All printers with stop codes that are preventing printing.
- Printers with Jobs – All printers with active job in the queue.



Print Management Console

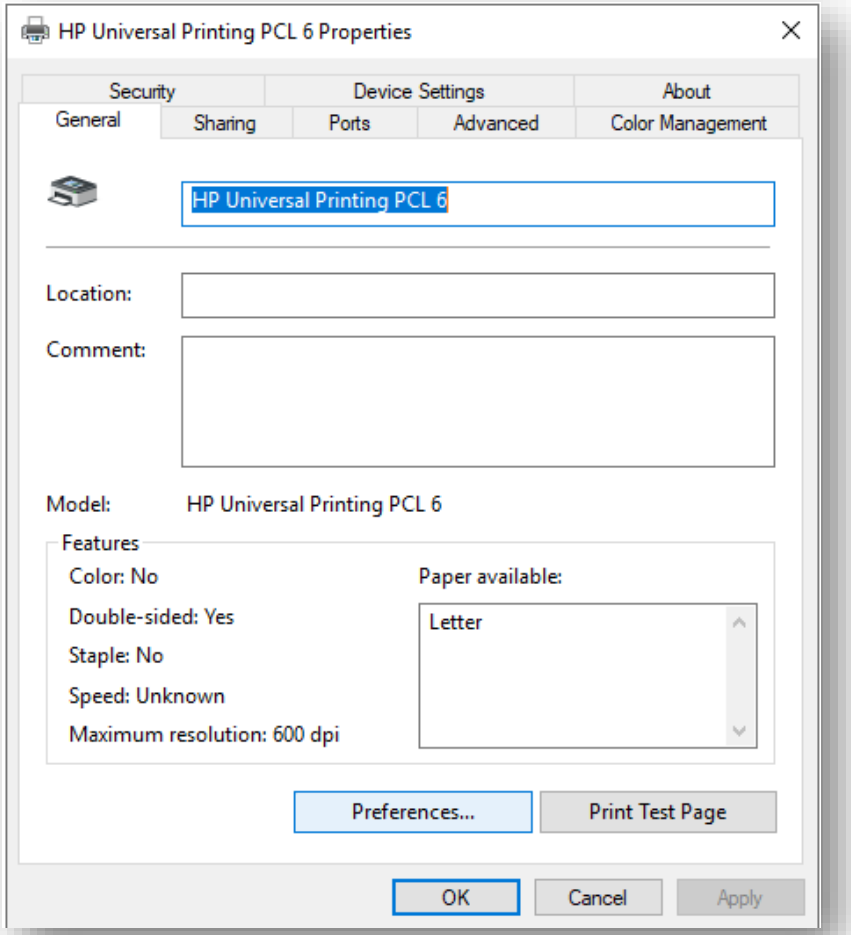
- Each print Server can be managed VIA the Print Management MMC snap-in.
- Drivers – manage all installed drivers for all printers (x86/x64).
- Forms – The paper specifications the printer can use to format print jobs.
- Ports – the local or TCP/IP ports that connect the printer to *this server.*
- Printers – The object that represents the Printer.



Print Properties vs Print Preferences

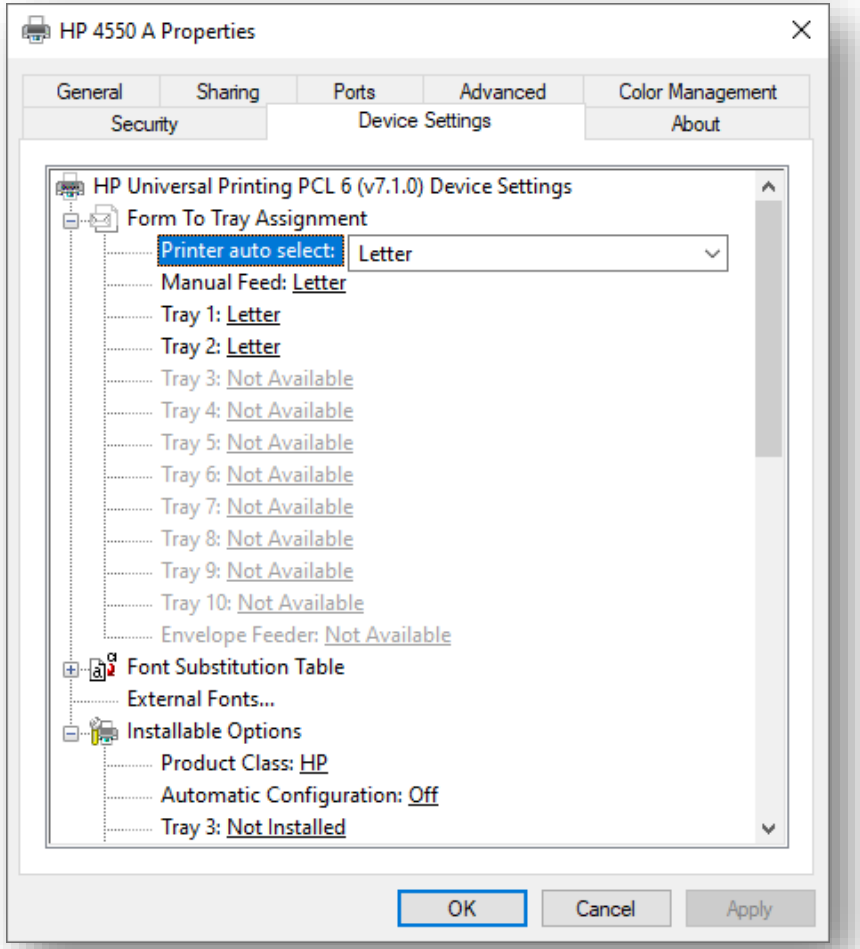
- Print Properties
  - Properties of the printer itself (how it’s connected, if it’s shared, physical tray settings)
- Print Preferences
  - Properties that apply to the *print documents themselves* (paper sizes, duplex printer, quality settings, watermarking).

Printer Properties General Tab



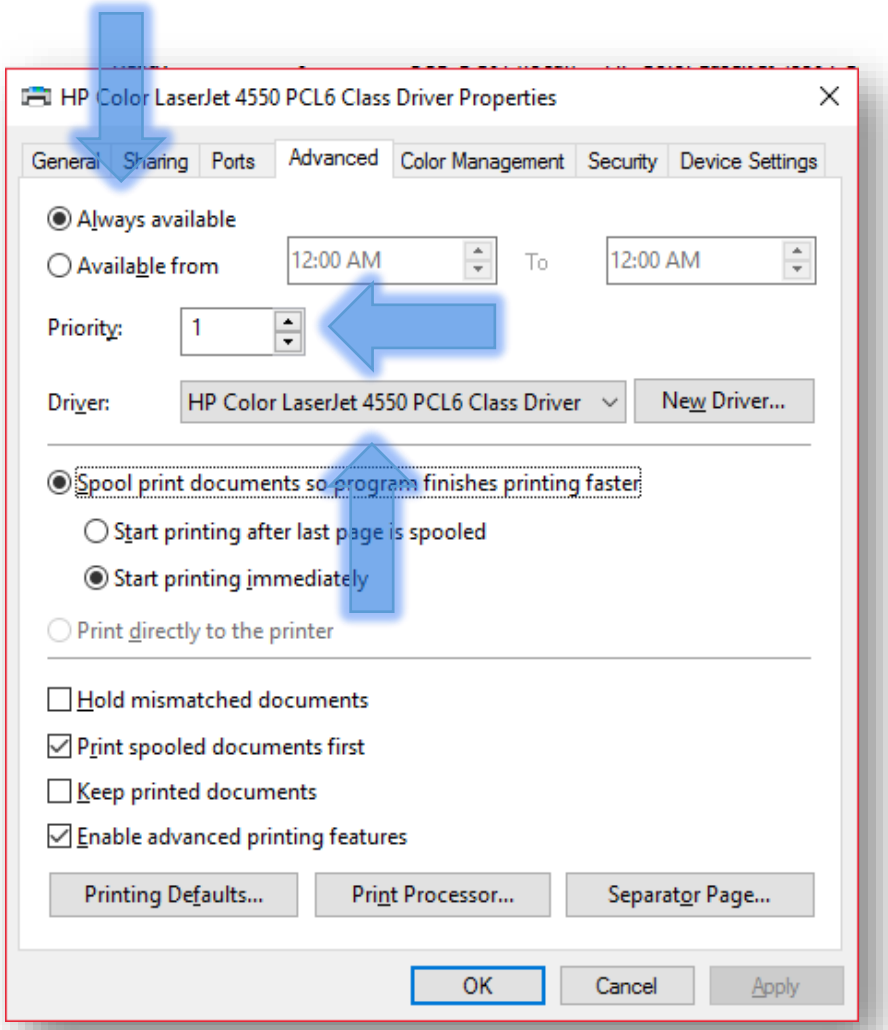
- The name and general demographic information can be found here.
- General Settings and capabilities.
- A button to go to print preferences
- Print Test Page button.

Printer Properties Device Settings tab



- Hardware options such as
  - duplexing
  - paper tray settings.
  - stappling
  - Per-tray configuration
  - collating

Printer Properties Advanced tab

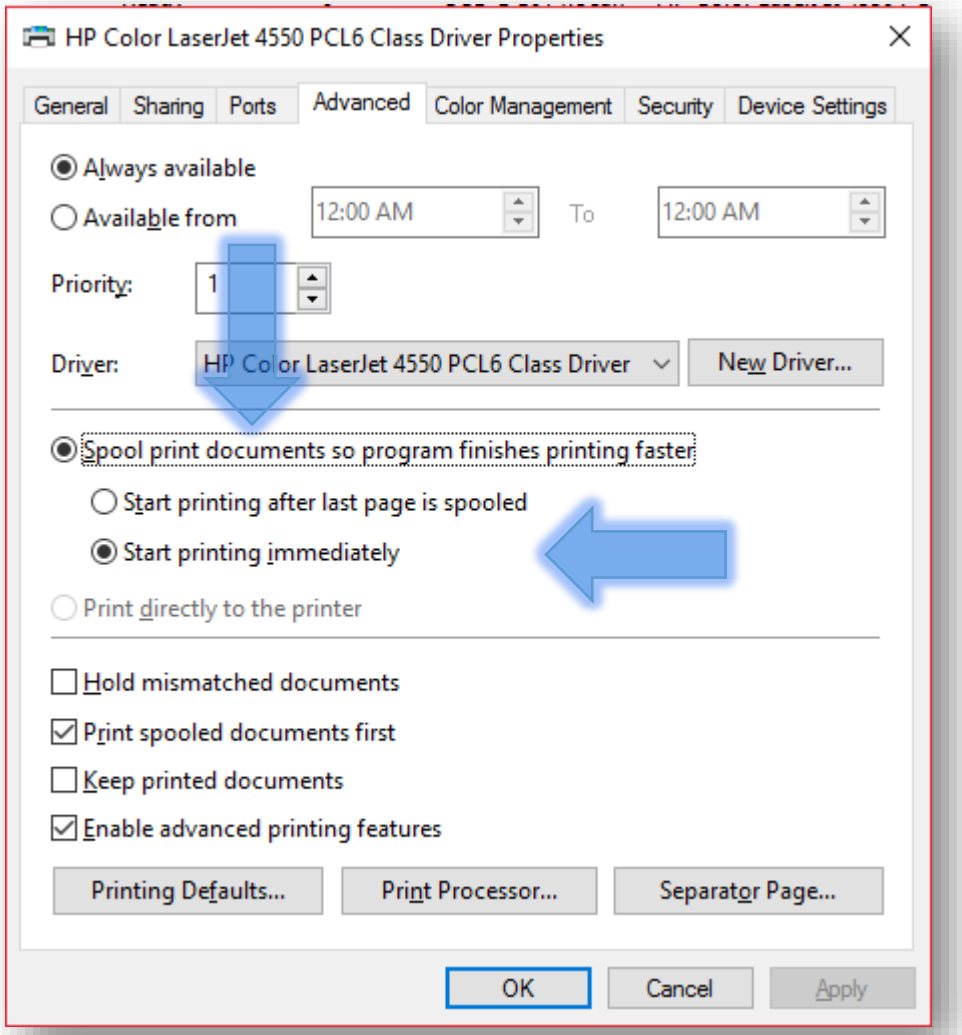


- **Availability**
  - Hour's printer is available, during off hours jobs **will be held until the printer comes online**
- **Priority**
  - Used when multiple **logical printers** are pointing to the *same physical* ***print device***. The printer with the **higher priority** setting will print first if two jobs arrive at the same time.
- **Driver**
  - Can be used to update the device driver used by the printer

Multiple Logical Printers for a Single Physical Print Device

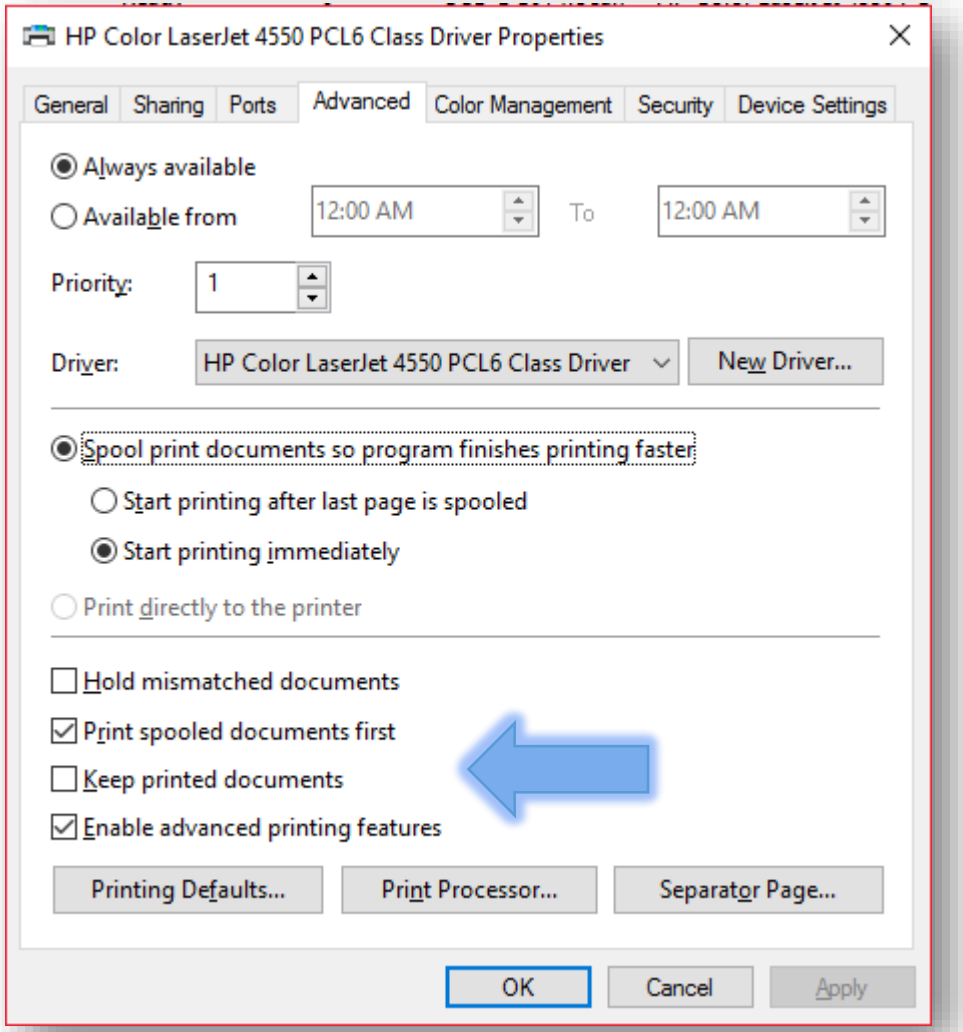
- Multiple logical printers connected to a *single physical printer* is a common method of applying multiple configurations to the same physical printer.
- Another common reason for creating multiple logical printers for a single physical printer, is to create priority levels for different print jobs. People who submit a print job to one logical printer have higher priority than people who submit to another.
  - To configure different priorities for the logical printers, select the Advanced tab of the printer's properties and specify a value for the Priority property in the range from 1 (lowest priority) to 99 (highest priority)

Printer Properties Advanced tab



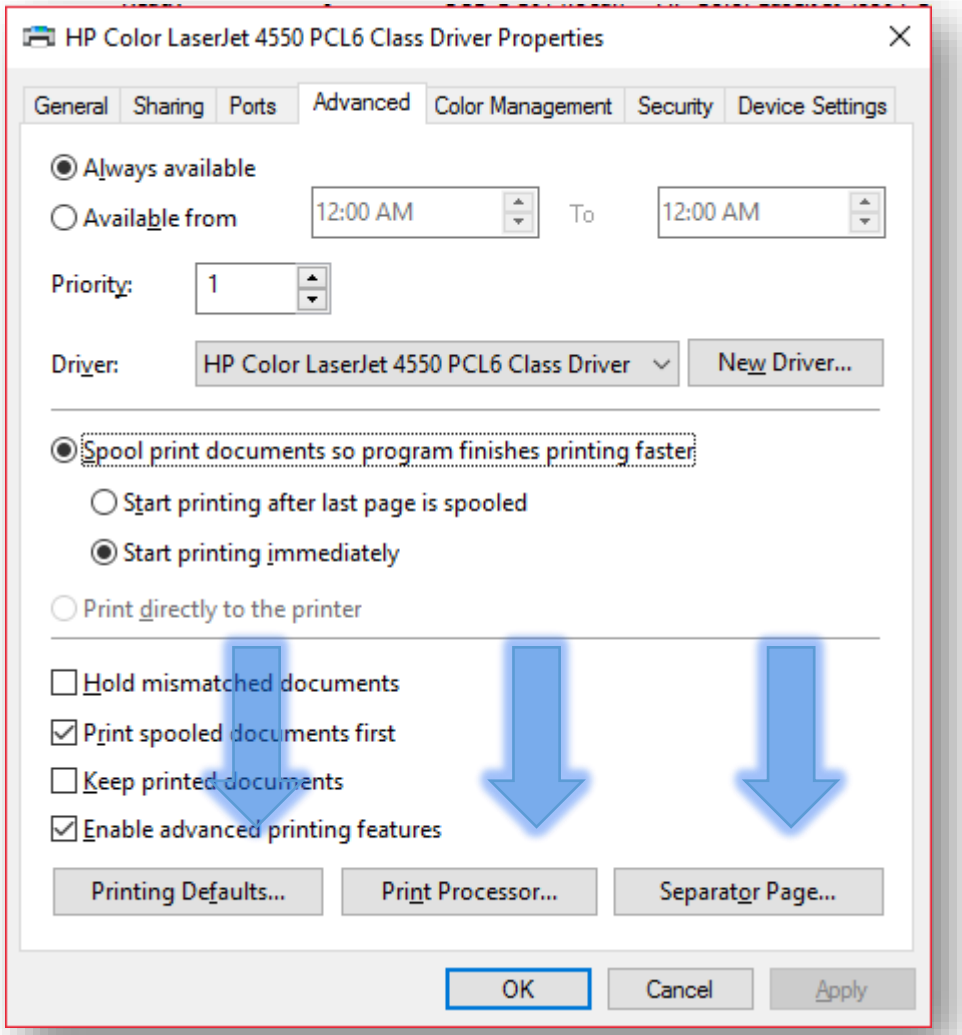
- Spooling configuration
  - Default Configuration: spool documents. Spool the printing job in the background; let the program continue to allow user interaction.
    - **Start Printing Immediately**: If the printer is configured to spool start printing the job as soon as the first page is spooled
    - **Start Printing after the last page is spooled**: wait until the entire job is spooled before printing.
  - Print directly to the printer: **do not spool the job**. This sends the job to the printer and continues to transmit and print simultaneously until it is finished. If this option is chosen, the print job must finish before the application can resume interacting with the user. This is usually much slower as well.

Printer Properties Advanced tab



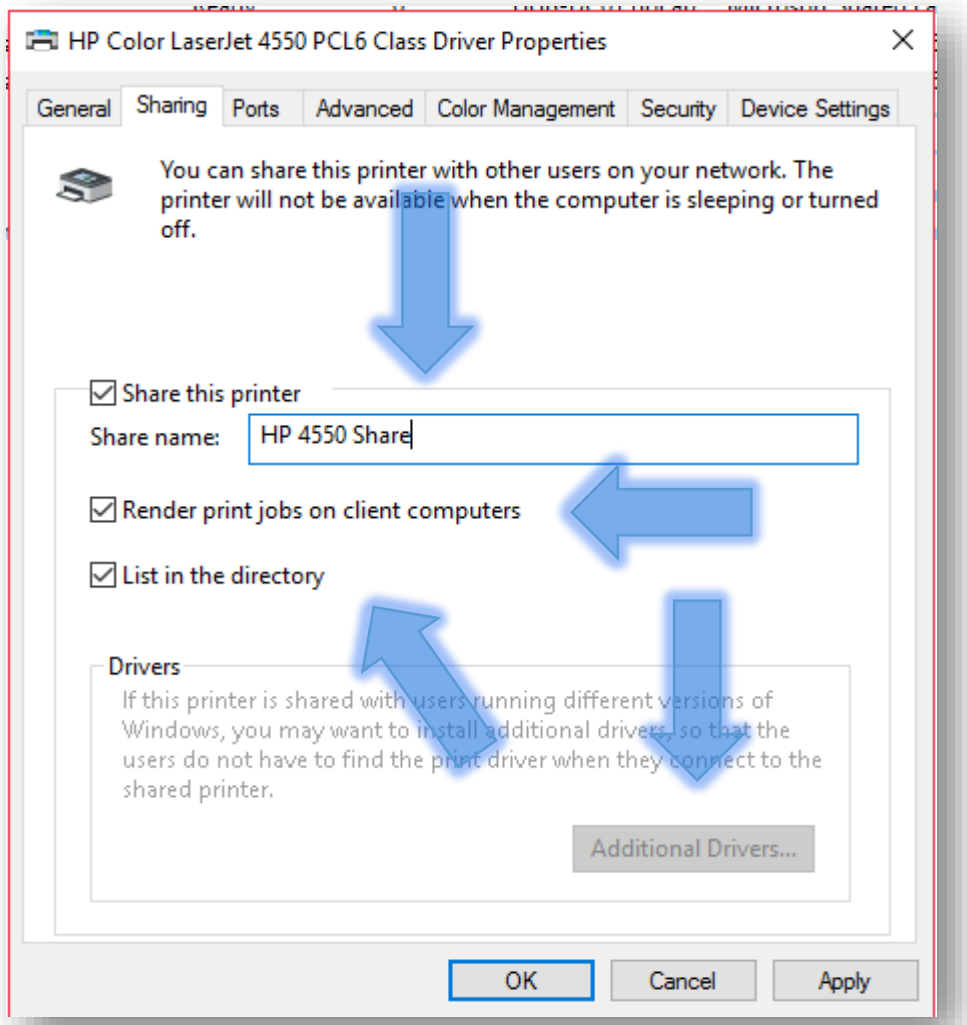
- **Hold Mismatched Documents**
  - Holds the print job in the queue if the job doesn't match the configuration of the printer (i.e. wrong paper size)
- **Print spooled documents first**
  - Priority is given to jobs that are finished spooling over jobs that have only started spooling (prevents large jobs from monopolizing the printer)
- **Keep printed documents**
  - Keeps a copy of the print job in the queue even after the job has finished printing
  - Allows jobs to be resubmitted later if the job fails to print or additional copies are needed
- **Enable advanced printing features**
  - Allows you to manipulate manufacturer-installed options on the printer that fall outside the default Windows printer settings

Printer Properties Advanced tab



- **Printing Defaults**
  - Specifies the default printer configuration options, such as whether jobs are duplexed, the orientation of the paper, and the quality of the print
- **Print Processor**
  - Allows you to choose the format of print jobs
- **Separator Page**
  - Allows you to specify a separator page that is inserted into the beginning of your print job. Useful so that people know which print job belongs to which user, but also a **giant waste of paper**

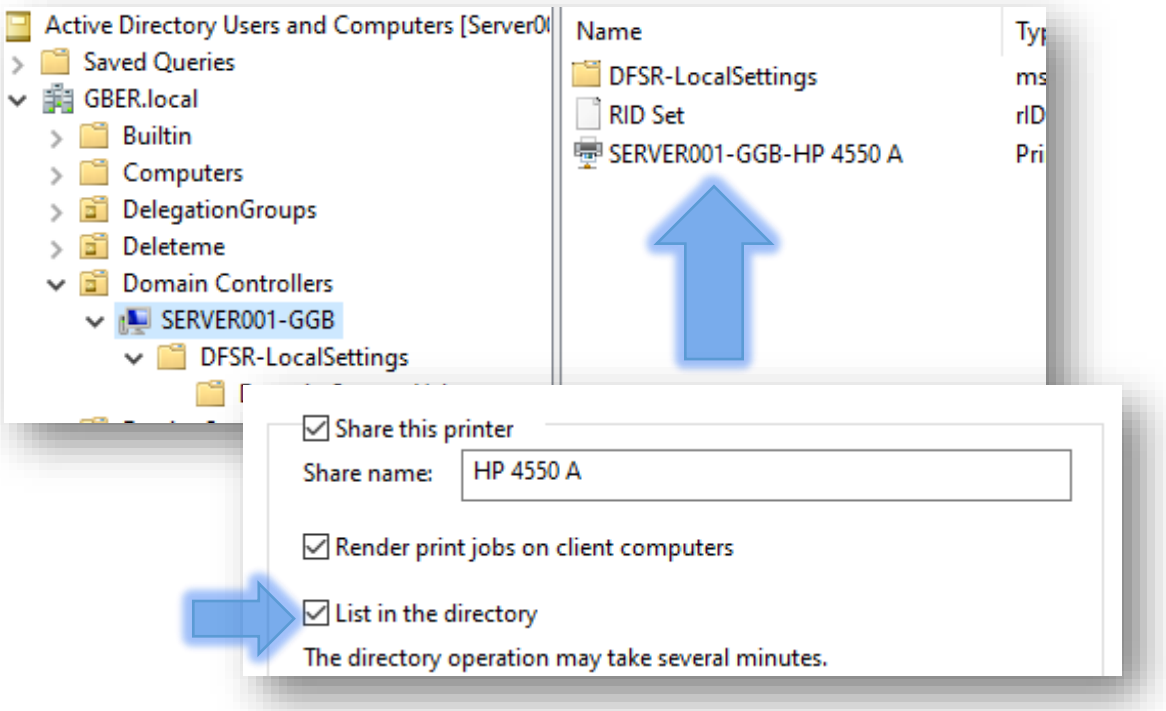
Printer Properties Sharing tab



- Render print jobs on client computers
  - Print job is only sent to the print server when the job has been spooled by the client's print spooler (offloads processing of the document to the client computer).
- List in the Directory
  - Make the printer a searchable object in AD.
- Additional Drivers
  - Windows clients will automatically download the printer driver from the print server when they connect to the printer.
  - Can be used to specify whether Itanium, x64 and/or x86 drivers will be stored on the print server for client systems with different processor architectures.

Printers Listed in AD

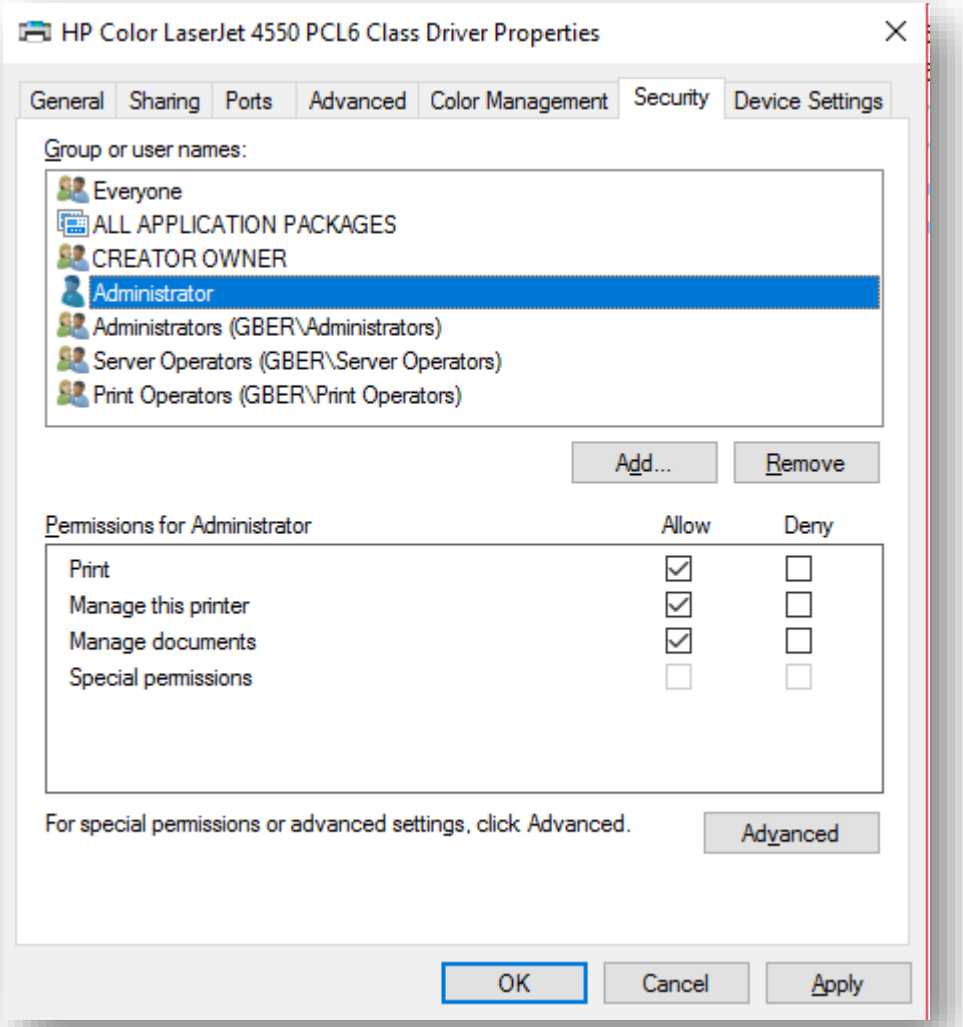
(and how to find them…)



- To see printers in ADUC, turn on View → Users, Contacts, Groups and Computers as containers.
  - The Printer share will be listed as a child of the Computer Object that's sharing it.
- Using Pwsh, and `Get-Printer`
  - Use the `-computerName` parameter and specify the Printer server's hostname.
  - Pipe to `Where-Object {$_.Published}`

```
Get-Printer -ComputerName Server1-GGB | Where-Object {$_.Published}
```

Printer Permissions



- Print
  - connect to a printer and submit print jobs to it.
  - pause, resume, restart and cancel the **user's own documents**
  - By default, this permission is assigned to Everyone
- Manage Printers
  - ***Implies*** Print permission
  - cancel **all** documents, share the printer, change printer properties, and change printer permissions
  - By default, Administrator have this permission
- Manage Documents
  - manipulation of the documents in the print queue by pausing, resuming, restarting, canceling, and reordering print job
  - cannot send documents or control the printer's status
  - assigned, by default, to the Creator Owner

Connecting to Shared Printers

There are several ways to connect a shared printer to another computers

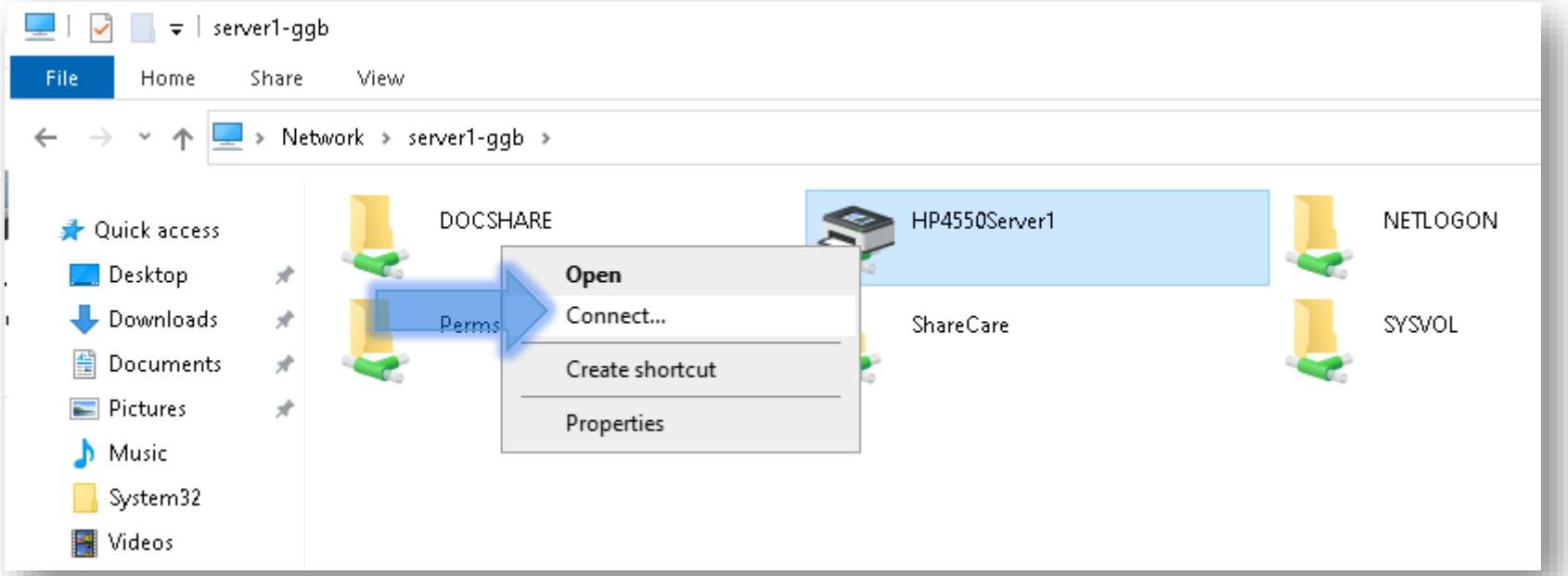
- Using Devices and printers control panel applet
- Using file explorer and the UNC path
- Using Powershell

Connecting to Shared Printers

- Using Devices and printers control panel applet
- Click Add Printer
  - Windows will search for ‘discoverable’ printers on the network and those published in AD.
- Click My printer wasn’t Listed
  - Add the printer by searching the directory (AD)
  - Add by UNC path
  - Add by TCP/IP
  - Add local printer (USB, LPT).

Connecting to Shared Printers

- Using file explorer and the UNC path
  - Open files explore and enter the UNC path to the printer or browse the network node for the Print Server.



Connecting to Shared Printers

- Using Powershell

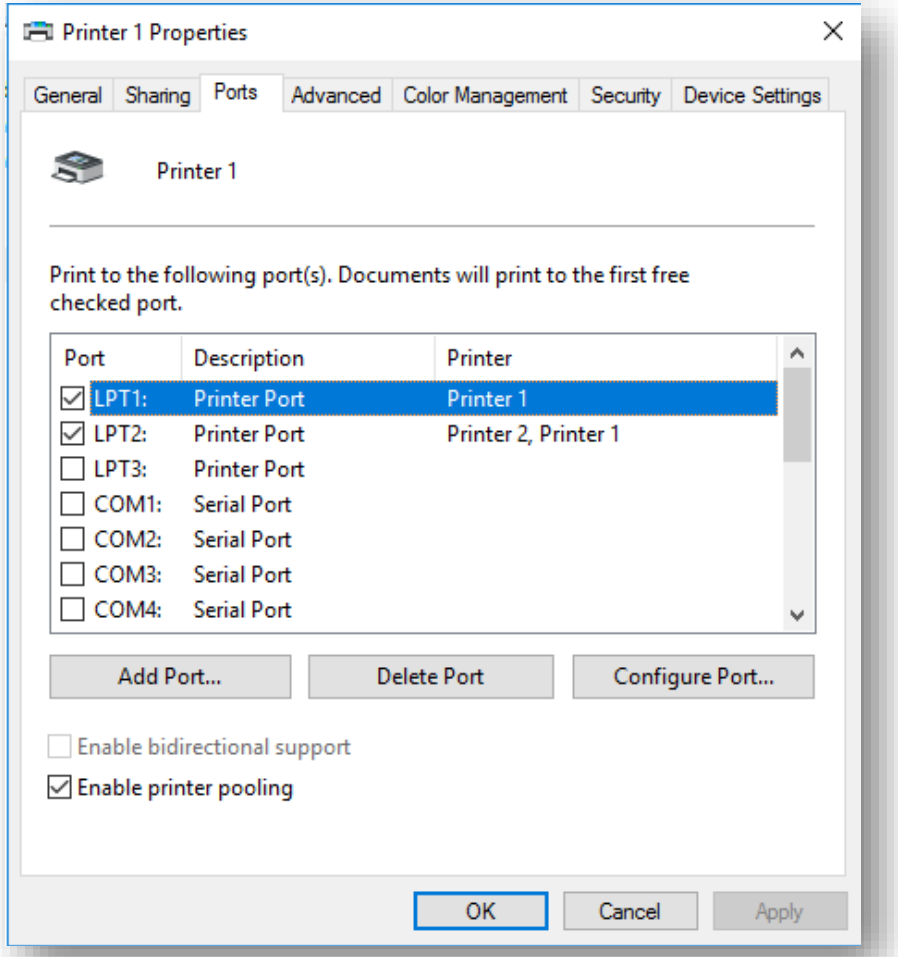
```
#Add the printer using a variable name
$printer = "HP 4550 B"
$prntUNC = “\\Server2-GGB\HP 4550 A"
Add-Printer -ConnectionName $prntUNC

#Show the printer
Get-Printer -Name $prntUNC

#remove the printer
Get-Printer –name $prntUNC | remove-Printer
```

Printer Pooling

- A **printer pool** is one logical printer that points to *multiple physical print devices*.
- The print server sends jobs to the first available physical print device or the next in the pool if the previous printer was occupied.
- Configured using the **Ports** tab of a printer's properties. If you select "**Enable Printer Pooling**", you can specify multiple ports containing print devices.

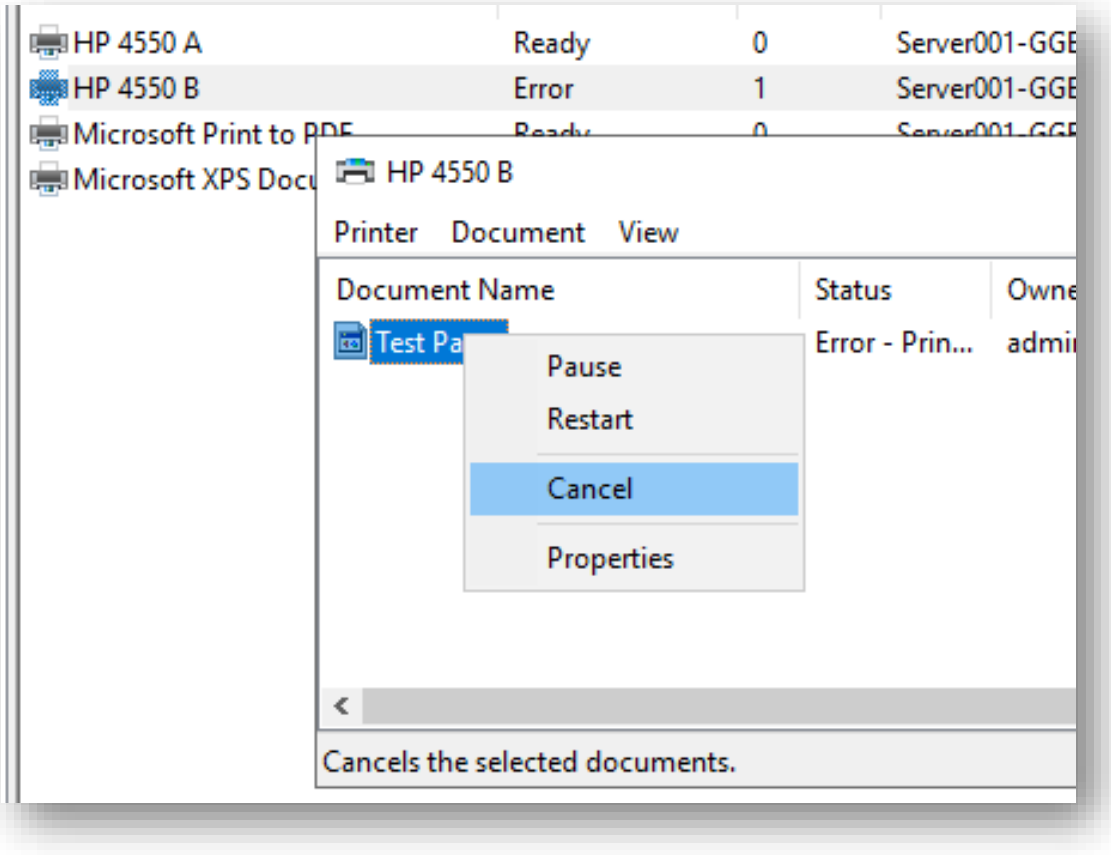


Do this…

- On Server2, install a second Printer to LPT2
- Name and Share name: HP4550 B
- Enable Print pooling on Printers A and B
- Print a Test page (After Pooling).

Solving Print Queue Problems

- One of the most common printing problems is **getting jobs stuck in the print queue**
- A stuck print job can **prevent other jobs from printing**
- When a job becomes stuck in the print queue, the first option to fix the problem is to **try canceling or restarting the job**, if that does not work, it sometimes helps to **restart the print spooler**



Solving Print Queue Problems

Removing Print Jobs with PWSH

```
#Printer name variable
$printer = “HP 4550 B”

#Check the print job queue
get-printjob -PrinterName $printer

Id    ComputerName    PrinterName     DocumentName         SubmittedTime        JobStatus
--    ------------    -----------     ------------         -------------        -----
3                     HP 4550 B       Test Page            3/20/2023 10:52:17 … 8210

#Remove the stuck job.  Use * to remove them all.
Remove-PrintJob -PrinterName $printer -ID 3
```

OR

```
Get-PrintJob -PrinterName $printer -ID 3 | Remove-PrintJob

Get-Printer | Get-PrintJob | Where-Object JobStatus -like '*error*’ | Remove-PrintJob
```

Restarting the Print Spooler

- Use the Services administrative tool and restart the Print Spooler service

  or

- Enter the commands
  - `net stop spooler` (followed by)
  - `net start spooler`

or

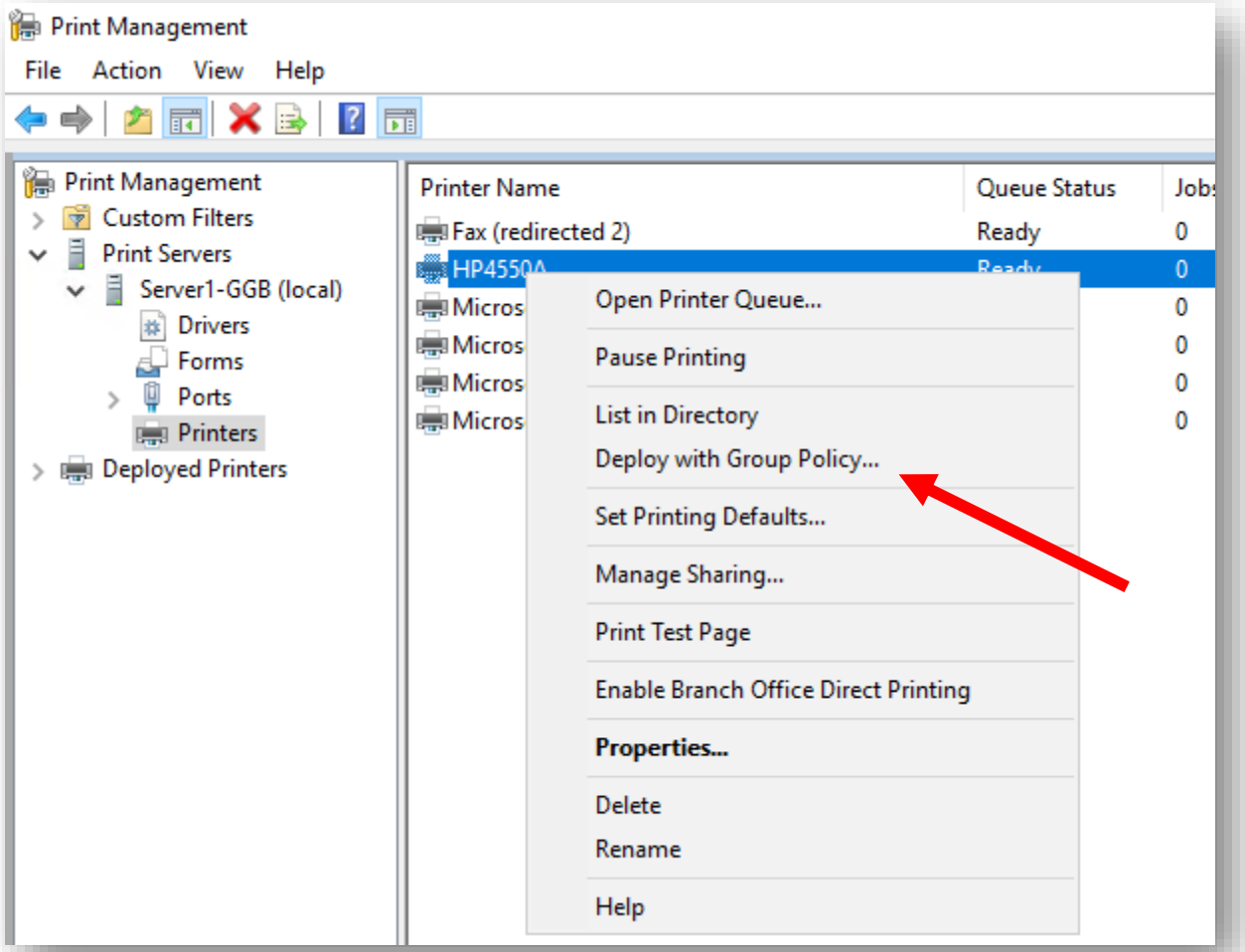
- Enter the cmdlet
  - `Stop-service spooler` (followed by)
  - `Start-Service Spooler`

*** Remember to reference the service name, not the common name***

Deploying Printers using Group Policy

- Using group policy, printers can be automatically assigned to users or computers to a Group Policy Object
- Printers are deployed to a GPO using the Print Management Console
- Can be deployed based on User, Computer or both.

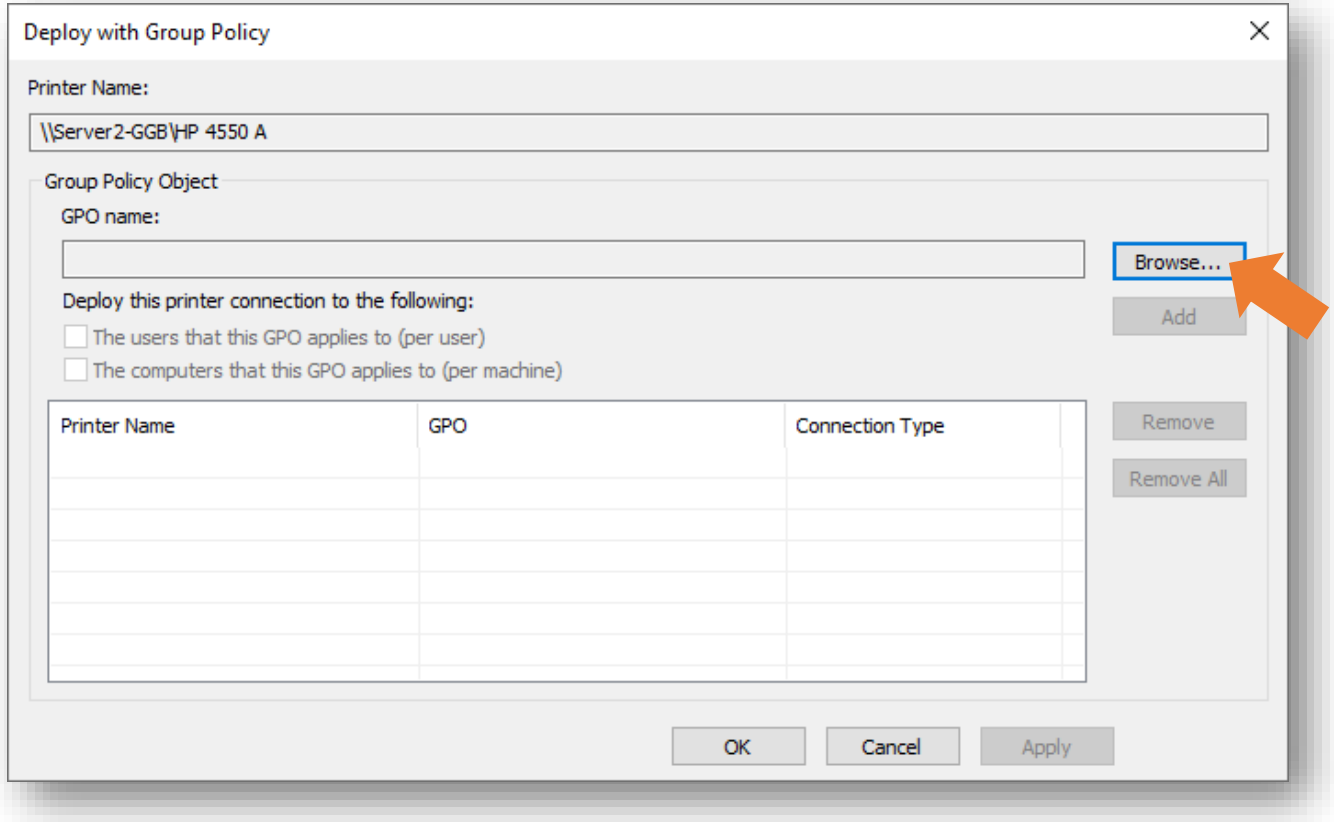
Deploying a Printer Using Group Policy



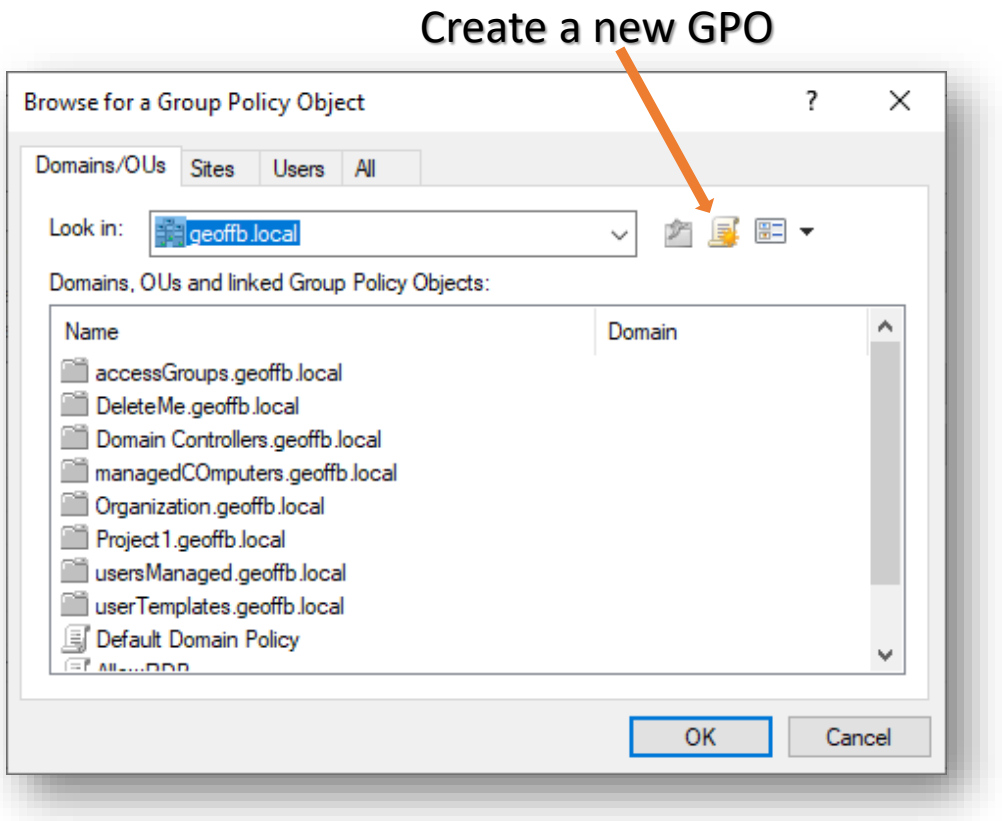
- Right-click the printer to deploy in Print Management, then choose the Deploy with Group Policy option

Deploying a Printer Using Group Policy

- Use the Browse button to select a GPO to add the deployed printer to (you can create a new GPO, if needed)

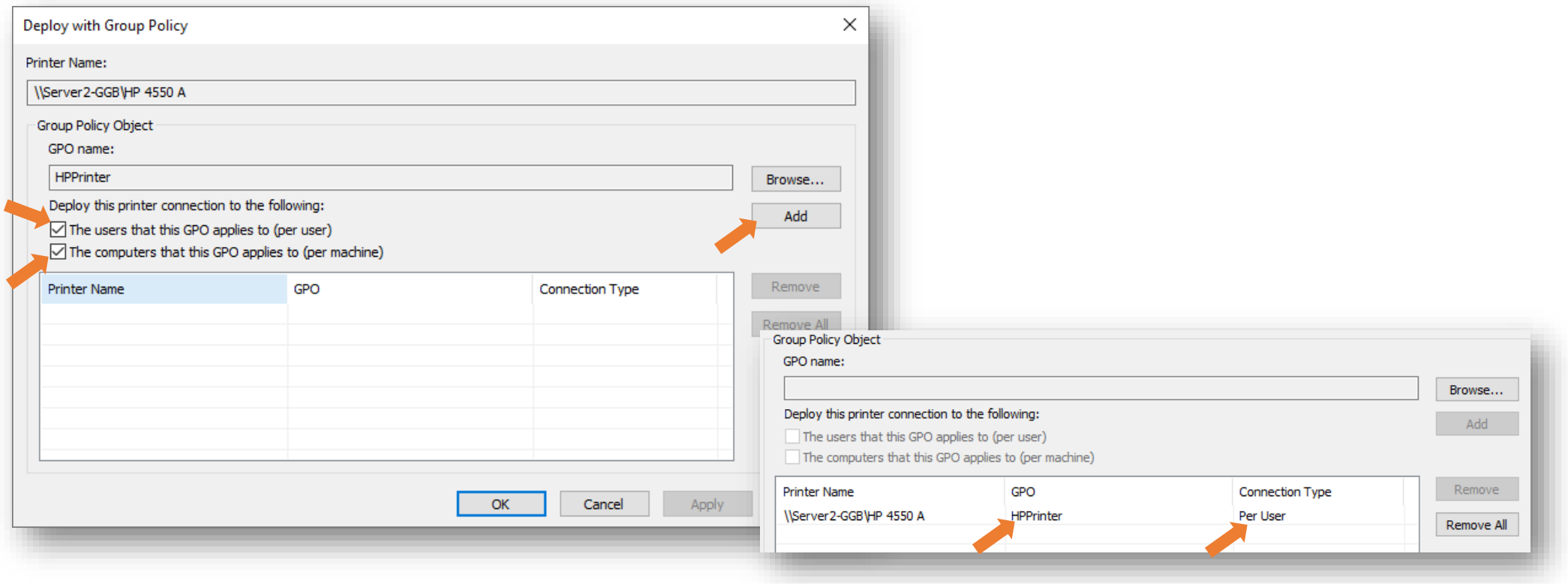


Deploying a Printer Using Group Policy



- The GPO should already be applied to the OU where the users or computers you want the printer deployed to

Deploying a Printer Using Group Policy



- Select whether the printer should be deployed to users or computers (or both)
- Be sure to click the Add button

Do this…

1. Open Print Mgmt console.
2. Select HP 4550 A Printer → Deploy with GP
3. Create a new GPO called PrintPolicy
4. Apply the Deployment Policy to Computers
5. Link the Policy to the managedComputers OU in GPMC.
6. Click Add